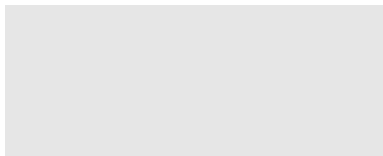
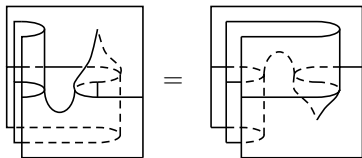
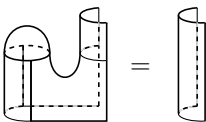


cuspidal relations



swallowtail relation



+ permutations

cusp flip relation

Morse relations